

B.M.S. COLLEGE OF ENGINEERING, BENGALURU
Autonomous Institute, Affiliated to VTU



**An AAT Report on
on**

“Automatic Classification of Sexism in Social Networks: An Empirical Study on Twitter Data”

**Submitted in fulfilment for the award of degree of
Master of Technology
In
Computer Science and Engineering**

Submitted by

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Under the Guidance of
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B.M.S. COLLEGE OF ENGINEERING
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**B.M.S. College of Engineering,
Department of Computer Science and Engineering**



DECLARATION

This is to declare that the AAT work entitled “Automatic Classification of Sexism in Social Networks: An Empirical Study on Twitter Data” carried out by P Yogesh Nayaka (1BM25SCS003-T), is a part of the course Artificial Intelligence (MCS101) of 1st Semester M.Tech (CSE), Department of Computer Science and Engineering, B.M.S. College of Engineering, Bangalore. This AAT work has been carried out under the guidance of Dr. Umadevi V during the academic semester Nov 2025 - Feb 2026. I declare that this work is original and has not been submitted elsewhere for any other degree or award.

Signature of the Candidate

P Yogesh Nayaka (1BM25SCS003-T)

B. M. S. COLLEGE OF ENGINEERING
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING



CERTIFICATE

Certified that the AAT entitled "Automatic Classification of Sexism in Social Networks: An Empirical Study on Twitter Data" is a bonafide work carried out by P Yogesh Nayaka (1BM25SCS003-T), in partial fulfillment for the award of Master of Technology in Computer Science and Engineering of the Visvesvaraya Technological University, Belagavi, during the academic year 2025-2026. This report satisfies the academic requirements prescribed for the said degree.

Guide

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1. Introduction:

Social Media has changed how people communicate, interact and create social identities. With its benefits, it also comes with drawbacks. People spread hate based on a person's colour, sex, nationality, political identity, religion etc. Sexism is among them, these comments are pervasive, since most people online are anonymous the problem is amplified. Exposure to these comments are not just unpleasant, they have consequences on mental health, public participation and freedom of expression.

Sexism is not always obvious. It is often subtle, normalized, objectification statements that are hidden as humor or concern. These expressions are widespread, socially embedded and overlooked by human moderators and automated systems.

2. Problem Statement:

Attempts are being made to reduce the number of comments posted online, most automatic systems are limited to identifying explicit hate speech or misogynist content. This approach is narrow and fails to capture the full spectrum of sexist behavior, particularly implicit and non-violent forms of sexism that hurt people's feelings while remaining socially normalized. This leaves small portions of comments to go unnoticed on social media.

This paper tackles the issue by introducing a new task for its detection, building a manually annotated dataset from Spanish Twitter posts, and testing both classical models and deep learning models.

3. Objectives:

The aim of the paper was to design and evaluate an automated approach for detecting sexist content on social media. It introduces a new sexism detection task, creating a manually annotated dataset of Twitter posts in Spanish and comparing the performance of classical machine learning models and deep learning models in identifying sexist posts. The study contributes towards the development of a more reliable and socially responsible content moderation system.

Though the objective of the paper is to explore automatic detection of sexist posts using machine learning models and deep learning models. The paper uses Twitter's API to pre-process the text data, due to unavailability of the original dataset, this

project adopts a dataset with texts in it rather than twitter_id. The work aims to preprocess raw text data, extract features and evaluate the performance of different classic machine learning models

4. Dataset Used:

The dataset used in this project is the Sexism Detection in English Texts dataset, it contains English text samples labeled for the presence of sexist content. The dataset has six columns rewire_id, text, label_sexist, label_category, label_vector and split. Table 1 given below will give detailed description on the dataset's attributes.

Column Names	Description	Data Type
rewire_id	Unique identifier for each text instance	String
text	English text sample	String
label_sexist	Binary label indicating sexist or non-sexist content.	Categorical
label_category	Type of sexism	Categorical
label_vector	Fine-grained sexism annotation	Categorical
split	Dataset split used for training or evaluation	Categorical

Table 1: Dataset Description

Each row has a unique identifier rewire_id which is a text segment. The text column has the tweets which are processed and used for training classic machine learning models. The label_sexist is the target column which has two classes: sexist and not sexist. With these columns the dataset also has label_category and label_vector, which describe the type of sexism. The dataset has a total of 20,000 rows, which are divided into training and testing sets. The training set has 16,000 rows in which 3,884 rows are labeled sexist and 12,116 rows labels are non-sexist. The testing set has 4,000 rows in

which 970 rows are labeled sexist and remaining 3,030 are labeled non-sexist. Table 2 below shows how train and test are split.

Class Label	Train	Test
Sexist	3,884	970
Not-Sexist	12,116	3,030

Table 2: Dataset Split

5. Tools Used for Implementation:

5.1 Programing Language:

- **Python:** Python was used due to its extensive support for machine learning libraries. Its simple syntax makes it easy for writing and experimenting with models.

5.2 Python Libraries:

- **Pandas:** Pandas is mainly used to work with structured data like tables. It offers DataFrame which is fast and flexible making it easy to clean, transform, filter, analyze etc datasets. Pandas is used to load, handle missing values, preprocess data and prepare data for training models.
- **Matplotlib:** Matplotlib is used for visualizing and plotting data in python. It offers a wide range of plots like bar chart, scatter plot, histogram, pie chart and subplots to analyze data visually and track model performance. Matplotlib is used to visualize evaluation metrics and compare models making it easy to interpret.
- **Seaborn:** Seaborn is a data visualization library in python built on top of matplotlib. It offers a high level interface for visualizing relationships within a dataset. It is useful for generating plots such as heatmaps etc, while offering better styling compared to matplotlib.
- **SciKit-Learn:** sklearn is used for building and evaluating machine learning models. Its offers common models like Simple Linear Regression, Logistic Regression, KNN, Decision Tree, Support Vector Machine, Random Forest etc, with mentioned models it also offers tools for feature extraction like TF-IDF etc and tools for evaluation like accuracy_score, classification_report etc.

5.3 Text Cleaning Libraries:

- **Regular Expression(re):** re is used for pattern matching and text cleaning. Mostly used for NLP tasks to remove problematic characters, URLs etc from text data. It standardizes raw text data before feature extraction.
- **String:** string is a helper function that helps in handling punctuations, digits and whitespaces.
- **Emoji:** emoji is used to detect, extract and remove emoji from text. Emojis adds noise to text data, this library cleans user-generated text by converting emojis to descriptive text.
- **NLTK:** nltk or Natural Language Toolkit offer a range of basic natural language processing tools. `nltk.corpus.stopwords` gives a list of common stop words that can be used to reduce noise in text. `nltk.stem.PorterStemmer` is used to reduce words to their root form, which groups similar words.

5.4 Machine Learning Models:

- **Logistic Regression:** Logistic Regression is a linear classification model that uses sigmoid function to find probability of a given data point.
- **Support Vector Classifier(SVC):** SVC uses a margin to classify points by finding optimal decision boundaries with maximum margin.
- **Random Forest Classifier:** Random forest is an ensemble model that uses multiple decision trees to improve accuracy.

Table 3 below mentions the softwares used and their version numbers.

Slnno	Software	Version Number
1	Python	3.12.12
2	Pandas	2.2.2
3	Numpy	2.0.2
4	NLTK	3.9.1
5	Matplotlib	3.10.0
6	RE	2.2.1
7	Seaborn	0.13.2
8	Emoji	2.15.0
9	Sci-kit Learn	1.6.1

Table 3: Softwares used and their versions

6. Github Link: [Twitter Text Classification](#)

7. Methodology:

This project follows the standard machine learning pipeline. First, loading and exploring the data. Pandas is used to understand the data and see class distribution. Then raw text was cleaned by lowercasing, removing punctuation, URLs, emojis, stopwords and stemming to reduce word to root form.

The data was split into training and testing split, the split made was 80/20. To prevent data leakage a pipeline is used with TF-IDF and machine learning models. The models chosen were Logistic Regression, SVC and Random Forest.

These models were trained on TF-IDF features in the pipeline. After training these models were evaluated using confusion matrix, accuracy, precision, recall, F1-score and ROC-AUC.

8. Results Obtained:

The trained models were evaluated on the test dataset using different metrics like confusion matrix, accuracy, precision, recall, F1-score and ROC-AUC.

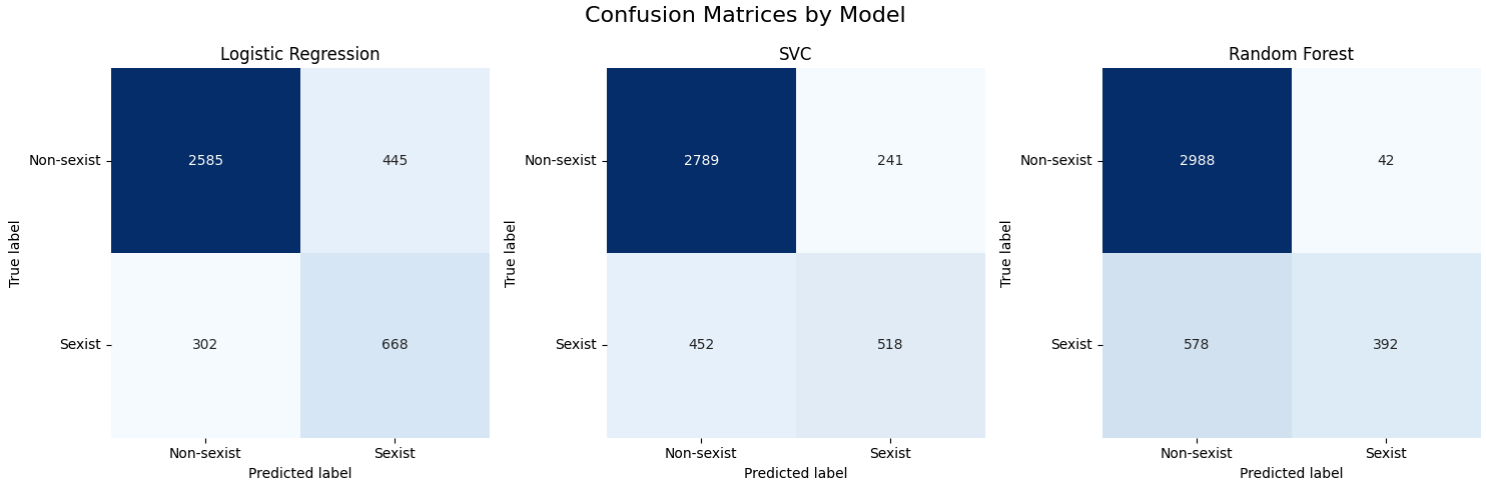


Figure 1: Confusion Matrix of each models

Figure 1 shows the confusion matrix of each model used. The confusion matrix shows that all three models have the same True Negative, but Logistic Regression has better True Positive and Random Forest is worse than the other models. Analysing further random forest, False Negative is high which is not ideal while detecting sexist content.

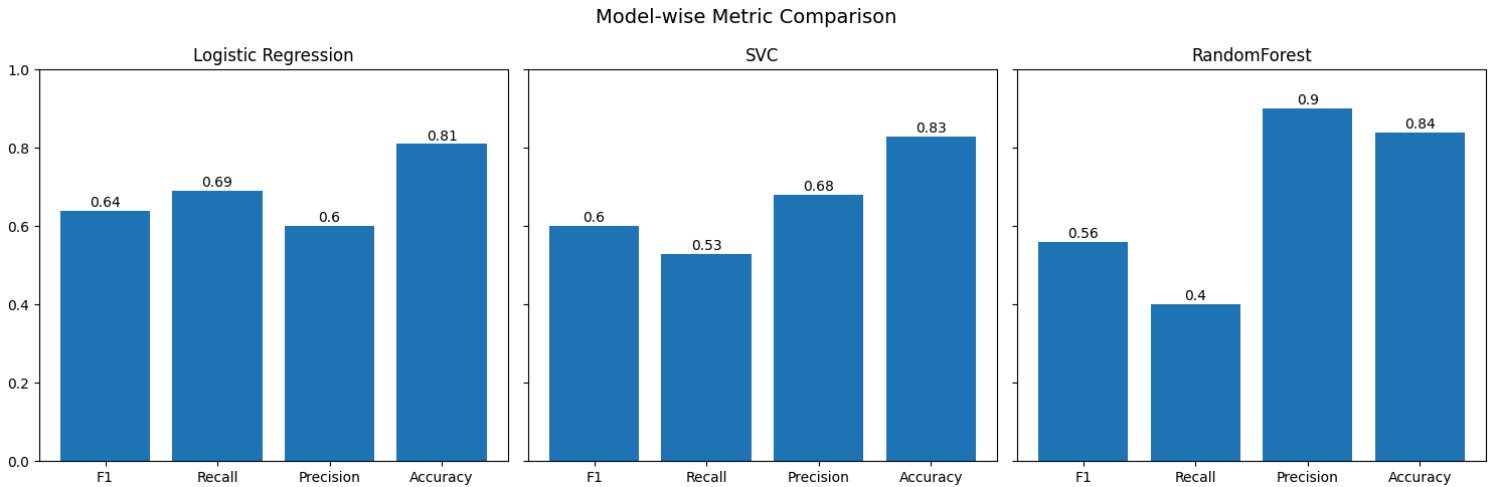


Figure 2: Metric Comparison of each model

Figure 2 shows metrics of each model used. Even though the accuracy of Random Forest is high compared to Logistic Regression and SVC, due to the highly imbalanced dataset and context of the dataset recall is used to catch all the sexist text.

Logistic Regression has recall of 0.69, SVC has recall of 0.53 and Random Forest has recall of 0.4. This suggests that Logistic Regression performs moderately on the given dataset. The lower values of SVC and Random Forest shows that the models miss large amounts of instances. Table 4 shows comparison of model metrics with those reported in the reference paper.

Model	Paper Result				Implemented Result			
	Accuracy	Precision	Recall	F1-Score	Accuracy	Precision	Recall	F1-Score
LR	0.71	0.62	0.64	0.62	0.81	0.6	0.69	0.64
SVC	0.71	0.61	0.63	0.61	0.83	0.68	0.53	0.60
RFC	0.72	0.67	0.57	0.6	0.84	0.9	0.4	0.56

Table 4: Comparison between paper and implementation

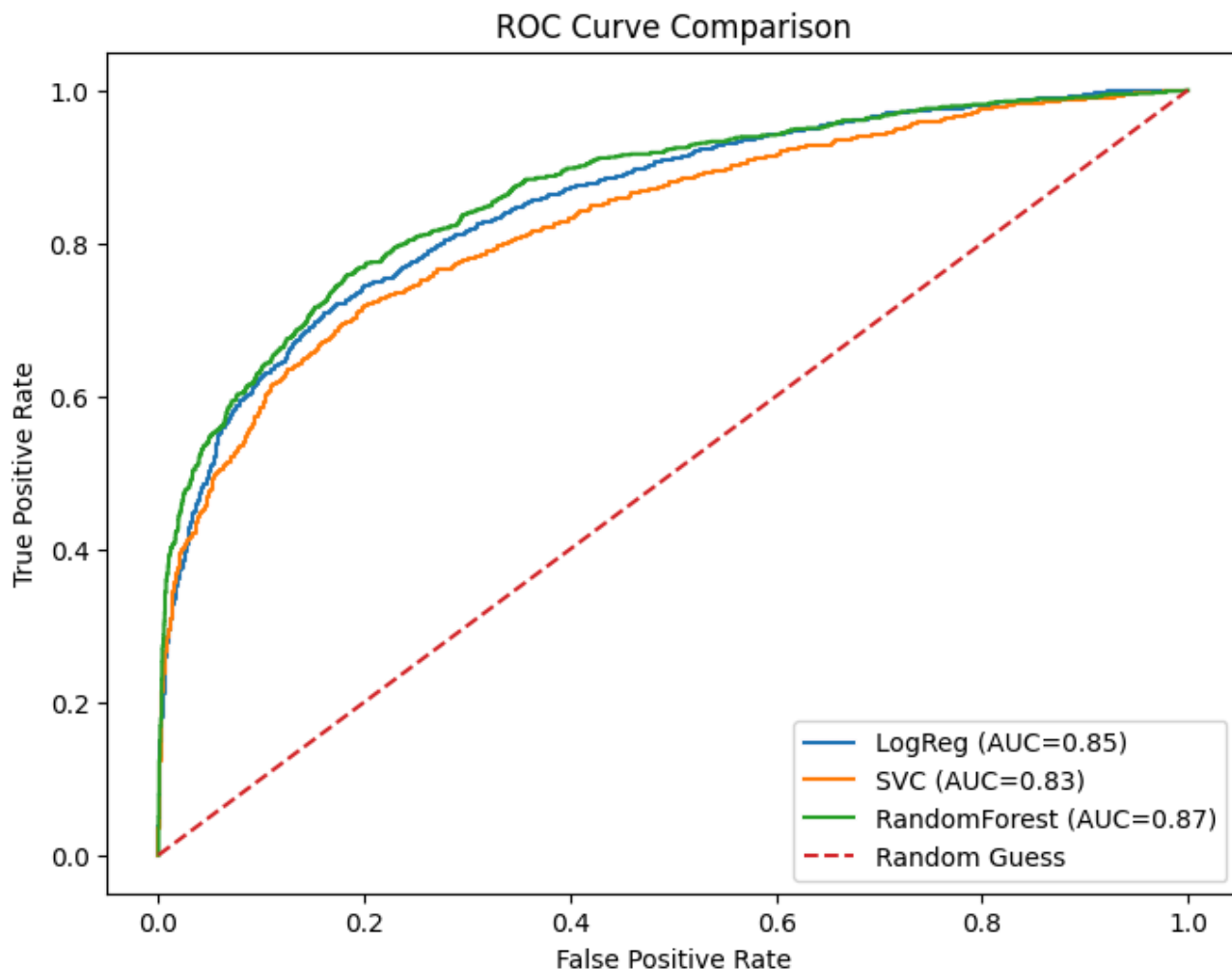


Figure 3: ROC-AUC of the models

Figure 3 shows ROC-AUC of the models. Random Forest, Logistic Regression and SVC has ROC-AUC of 0.87, 0.85 and 0.83 respectively. Although Random Forest has the highest ROC-AUC, its low recall was low for sexist content. Even though Logistic Regression was slightly low ROC-AUC, it achieved better recall and F1-score for sexist class.

9. Learning Outcome:

Through this project, a practical understanding of nlp and text classification was developed. Highlights include text preprocessing, feature extraction using TF-IDF and model selection for respective tasks. The project also provided hands-on experience with evaluating machine learning models using different metrics, not just accuracy.

This project helped understanding of applying machine learning techniques to real-world social issues like sexism detection in text data.

10. Paper Reference:

- Rodríguez-Sánchez, F., Carrillo-de-Albornoz, J. and Plaza, L., 2020. Automatic classification of sexism in social networks: An empirical study on twitter data. IEEE Access, 8, pp.219563-219576.

11. Screenshot of Similarity & AI Generated report: